
Your Research Title Here

Your Name

Your Institution

June 18, 2026

1 Problem Statement

The main problem is that parking at Wits is unpredictable and inconvenient. There is a real chance you will not find parking near the building you need to reach, and there is no way to know this without driving to the lot and searching for a space that may not exist.

1.1 Issues to Address When Developing a Solution

1.1.1 Detection

How do we detect whether a car occupies a given parking space?

1.1.2 Data Production

What kind of data does our chosen detection method produce?

1.1.3 Data Collection

How do we collect data from the sensors, where do we store it, and how do we process it?

1.1.4 Presentation

How do we present the data to the public?

1.1.5 Extra Problems

Bikes are a possible complication. They occupy parking spaces differently from cars and can introduce unexpected variables into detection and counting.

2 Pre-Solved Solutions

This section surveys existing approaches, frameworks, tools, or methods that have been applied in the wild or in prior research to address the problems identified above.

2.1 Detection

2.1.1 Ultrasonic Sensors

Ultrasonic sensors detect whether a car occupies a space by sending out sound waves and measuring how long they take to return. A car in the space changes that return time. The sensors cost around R25 each. The most straightforward implementation is to place one sensor per parking space and have it report whether the space is occupied.

Other solutions and studies include:

- https://github.com/CarlosFULLHD/iot_garaje_inteligente
- <https://github.com/aaryaapg/Smart-Parking>
- <https://github.com/RahulN25/Smart-Parking-System>
- <https://github.com/MahmoudHanyFathalla/Car-parking-system>
- <https://www.hackster.io/techgyanset/esp32-smart-parking-detection-system-d68660>

2.1.2 Infrared Sensor

Infrared sensors work similarly to ultrasonic sensors but use light instead of sound. Standalone transmitters or receivers cost between R4 and R20 each, depending on whether you buy individual components or a complete module. You can implement them by placing one sensor per parking space to detect occupancy, or by mounting them at the lot entrance to count vehicles as they pass.

Solutions and studies include:

- <https://github.com/aaryaapg/Smart-Parking>
- <https://github.com/RahulN25/Smart-Parking-System>
- <https://github.com/vishnubv944/smartParkingSystem>
- <https://github.com/prince61299/Smart-Car-Parking-using-ESP32>
- <https://github.com/harshad8782/Smart-Parking-System-IoT>
- <https://github.com/ferasaljoudi/AljoudiParkingSystem>

2.1.3 Magnetometers

Magnetometers detect the strength, direction, and relative change of magnetic fields. Vehicles produce a detectable disturbance in the earth's magnetic field, so a magnetometer

embedded in or near each parking space can determine occupancy. These modules cost around R65 each.

Solutions and studies include:

- https://github.com/ColoradoSchoolOfMines/parking_sensor
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11721214>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12437643>
- <https://patents.google.com/patent/US20220076576A1/en>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11322028>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10991249>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10332398>
- <https://www.mokosmart.com/lorawan-wireless-vehicle-detection-sensor-lw009/>
- <https://www.thethingsnetwork.org/device-repository/devices/bosch/tps110/>
- <https://www.thethingsnetwork.org/device-repository/devices/nwave/nps310sm/>
- <https://www.macnman.com/lorawan/sensors/lorawan-parking-sensor-paknode-gen-one>
- <https://www.chinaiotdevices.com/iotdevices/ultrasonic-and-geomagnetic-parking-occu>

2.1.4 Computer Vision

Computer vision uses a camera and a machine learning algorithm to determine whether a parking space is occupied. Camera sensor modules cost around R62 each, and e-waste such as old phones is another option. You mount the camera in a position where it can see the majority of the parking lot, and the algorithm interprets the feed in real time.

Solutions and studies include:

- <https://github.com/aswin-sreekumar/Smart-parking-system>
- <https://github.com/RahulN25/Smart-Parking-System>
- <https://patents.google.com/patent/US11182598B2/en>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/7893847>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12067878>

2.1.5 LiDAR

LiDAR measures the distance to objects by firing laser pulses and recording how long the reflection takes to return. LiDAR sensor modules range from R100 to R3000. You

implement them by pointing the sensor at each parking space to detect whether a vehicle is present.

Solutions and studies include:

- https://projecthub.arduino.cc/noah_barkol/arduino-nano-esp32-fully-automatic-garag

2.1.6 Pressure Sensors / Load Cell

A pressure sensor converts an applied force into an electrical signal. Basic modules cost just over R20 each, while more capable versions cost around R120. You implement them by placing a sensor at each parking space; the weight of a vehicle triggers the sensor and marks the space as occupied.

Other solutions include:

- <https://github.com/ferasaljoudi/AljoudiParkingSystem>

2.1.7 NFC / RFID

RFID uses radio waves to automatically identify and track tagged objects. A sensor and tag module costs around R285, with longer-range versions (up to 40cm) costing close to R400. You implement them at the lot entrance to count how many cars enter and leave.

Other solutions include:

- <https://github.com/Muhaiminul-Hasan/RFID-based-Smart-Parking-System-using-ESP32>
- <https://github.com/emregulerr/IoT-Smart-Parking-System>
- <https://github.com/anmoljhamb/smart-parking-system>

2.1.8 Radar

Radar detects the presence of objects using radio waves. Radar sensors start at around R70 each. You implement them by pointing a sensor at each parking space to detect whether a vehicle is present.

Other solutions include:

- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10263461>

2.2 Data Production

2.2.1 A Counter That Counts How Many Cars Enter and Leave

This approach tracks how many cars enter and leave a parking lot relative to the total number of spaces. The typical sensors for this are IR and RFID.

Solutions and studies include:

- <https://github.com/vishnubv944/smartParkingSystem>
- <https://github.com/prince61299/Smart-Car-Parking-using-ESP32>

- <https://github.com/Muhaiminul-Hasan/RFID-based-Smart-Parking-System-using-ESP32>
- <https://github.com/emregulerr/IoT-Smart-Parking-System>
- <https://github.com/anmoljhamb/smart-parking-system>
- <https://github.com/ferasaljoudi/AljoudiParkingSystem>

2.2.2 Check Each Parking Spot for a Car

This approach checks each individual parking space and determines whether it is occupied. The sensors used are proximity-based: IR, ultrasonic, pressure, LiDAR, computer vision, and magnetometers. You either place a sensor on each space or position one sensor to monitor a group of spaces.

Solutions and studies include:

- <https://github.com/aaryaapg/Smart-Parking>
- <https://github.com/RahulN25/Smart-Parking-System>
- https://github.com/Dipal018/Smart_Parking
- https://github.com/CarlosFULLHD/iot_garaje_inteligente
- <https://github.com/harshad8782/Smart-Parking-System-IoT>
- <https://github.com/aswin-sreekumar/Smart-parking-system>
- <https://www.hackster.io/techgyanset/esp32-smart-parking-detection-system-d68660>
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- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11322028>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10991249>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10332398>
- <https://patents.google.com/patent/US11182598B2/en>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/7893847>
- <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12067878>
- <https://www.thethingsnetwork.org/device-repository/devices/bosch/tps110/>

2.3 Data Collection

2.3.1 WiFi to Server

A centralised server stores the parking data. A microcontroller on site sends live readings to the server, which handles computation and makes the data available to other systems.

Solutions and studies include:

- <https://github.com/aaryaapg/Smart-Parking>
- <https://github.com/RahulN25/Smart-Parking-System>
- <https://github.com/vishnubv944/smartParkingSystem>
- https://github.com/Dipal018/Smart_Parking
- <https://github.com/prince61299/Smart-Car-Parking-using-ESP32>
- <https://github.com/harshad8782/Smart-Parking-System-IoT>
- <https://github.com/emregulerr/IoT-Smart-Parking-System>
- <https://github.com/aswin-sreekumar/Smart-parking-system>
- <https://www.hackster.io/techgyanset/esp32-smart-parking-detection-system-d68660>
- <https://www.thethingsnetwork.org/device-repository/devices/bosch/tps110/>

2.3.2 Local Microcontroller

The microcontroller handles computation locally and presents the data on site without relying on an external server.

Solutions include:

- https://github.com/Dipal018/Smart_Parking
- https://github.com/CarlosFULLHD/iot_garaje_inteligente
- <https://github.com/Muhaiminul-Hasan/RFID-based-Smart-Parking-System-using-ESP32>
- https://github.com/ColoradoSchoolOfMines/parking_sensor
- https://github.com/elise-ng/FYP_SmartCarPark

2.4 Presentation

2.4.1 App

A mobile app gives users a live view of each parking lot at any given time.

Solutions and studies include:

- <https://github.com/aaryaapg/Smart-Parking>

- <https://www.hackster.io/techgyanset/esp32-smart-parking-detection-system-d68660>

2.4.2 Website

A website serves the same function as the app but runs in a browser.

Solutions and studies include:

- https://github.com/Dipal018/Smart_Parking
- <https://github.com/vishnubv944/smartParkingSystem>
- https://github.com/CarlosFULLHD/iot_garaje_inteligente
- <https://github.com/prince61299/Smart-Car-Parking-using-ESP32>
- <https://github.com/Muhaiminul-Hasan/RFID-based-Smart-Parking-System-using-ESP32>
- <https://github.com/harshad8782/Smart-Parking-System-IoT>
- <https://github.com/emregulerr/IoT-Smart-Parking-System>
- https://github.com/ColoradoSchoolOfMines/parking_sensor
- <https://github.com/aswin-sreekumar/Smart-parking-system>
- https://github.com/elise-ng/FYP_SmartCarPark
- <https://www.thethingsnetwork.org/device-repository/devices/bosch/tps110/>
- <https://github.com/api-evangelist/spothero>
- <https://github.com/janmattfeld/parkme>

2.4.3 Live Counter

A physical display shows how many parking spaces are currently available.

Solutions include:

- <https://github.com/ferasaljoudi/AljoudiParkingSystem>

2.4.4 LEDs Above Parking Spot

An LED above each parking space indicates whether that space is free or occupied.

Solutions include:

- https://github.com/Dipal018/Smart_Parking
- <https://www.hackster.io/techgyanset/esp32-smart-parking-detection-system-d68660>

3 Strengths and Limitations of Each Solution

A critical assessment of each solution identified in Section 2, examining what works well and where each approach falls short.

3.1 Evaluation: Sensors

3.1.1 Ultrasonic Sensor

Strengths:

- Small and unnoticeable.
- Cheap per piece.

Limitations:

- Needs one sensor per parking space.
- Prone to dirt and debris blocking the sensor face.
- Cannot distinguish between a car and another object of similar size.

3.1.2 IR Sensor

Strengths:

- Small and unnoticeable.
- The cheapest option per piece.
- Depending on the application, one sensor per space is enough.

Limitations:

- Prone to dirt blocking the beam.
- Some applications require multiple sensors spread across the lot.
- Cannot distinguish between a car and other objects.

3.1.3 Magnetometers

Strengths:

- Detects cars specifically, since vehicles produce a measurable magnetic field disturbance that most other objects do not.
- Unaffected by dirt, weather, or obstructions since the sensor does not rely on line of sight.

Limitations:

- Costs more per piece than IR or ultrasonic sensors.

- Requires calibration for each installation site, since local magnetic field baselines differ.

3.1.4 Computer Vision

Strengths:

- Can monitor the entire parking lot at once.
- Only one unit is needed per lot.
- Can distinguish between cars and other objects.

Limitations:

- Complex to implement.
- Expensive per unit.
- Any obstruction of the camera eliminates detection entirely.

3.1.5 LiDAR

Strengths:

- Very high accuracy.
- Depending on the chosen sensor, one unit can cover multiple spaces.

Limitations:

- Expensive per piece, with costs reaching R3000 for higher-end sensors.

3.1.6 Pressure Sensor

Strengths:

- One sensor per space is enough in most applications.
- Can distinguish between cars and lighter objects based on weight thresholds.
- Not affected by dirt or anything covering the sensor surface.

Limitations:

- More expensive per piece than IR or ultrasonic sensors.
- Requires installation at each individual parking space.

3.1.7 RFID

Strengths:

- RFID is already built into Wits student cards, reducing the need for additional tags.
- Depending on the application, one reader at the lot entrance is enough.
- Passive RFID tags do not need an external power source.

Limitations:

- Longer-range readers can be expensive.

3.1.8 Radar**Strengths:**

- Works in all weather conditions since radio waves are not affected by rain, fog, or darkness.
- Can cover a wider area than most point-based sensors, reducing the number of units needed.

Limitations:

- Can produce false positives if nearby metal objects or moving people enter the detection range.
- More complex to configure and tune accurately than simpler proximity sensors.

3.2 Evaluation: Data Production**3.2.1 A Counter That Counts How Many Cars Enter and Leave****Strengths:**

- Only one sensor is needed at the entrance.
- Simple to implement.

Limitations:

- Cannot tell you which specific spaces are available, only the total count.

3.2.2 Check Each Parking Spot for a Car**Strengths:**

- Tells you exactly which spaces are available.

Limitations:

- More expensive due to the number of sensors required.
- More complex to implement and maintain.

3.3 Evaluation: Data Collection**3.3.1 WiFi to Server****Strengths:**

- Makes it easy to feed data into an app or website.
- Consolidates data from multiple lots in one place.

Limitations:

- Requires a server running continuously in the background.
- Can be expensive to set up and maintain.

3.3.2 Local Microcontroller

Strengths:

- Straightforward to implement for simple setups.
- Operates independently without relying on external infrastructure.

Limitations:

- Harder to feed the data into an app or website.

3.4 Evaluation: Presentation

3.4.1 App

Strengths:

- Shows live data for all lots at once.
- Supports push notifications.
- Navigation to available spaces is possible.

Limitations:

- Users need to download and install the app before they can use it.

3.4.2 Website

Strengths:

- No download required.
- Navigation to available spaces is possible.
- Shows live data for all lots at once.

Limitations:

- Requires an internet connection.
- Users need to open a browser and navigate to the link.

3.4.3 Live Counter

Strengths:

- Simple to implement.
- Cheaper than app or website solutions.

Limitations:

- Does not tell you which specific spaces are free.

3.4.4 LEDs Above Parking Spot

Strengths:

- When paired with a live counter, fills in the gap of not knowing which specific spaces are free.

Limitations:

- Only useful once you are already inside the lot and close to the space itself.

4 stuff to do

- search on design thinking
- mainly look at the problem space specifically the user persona
- for presentation look at both problem and solution space